

Freight Rate Prediction

Ahmad Fraij

1 Problem and Data

The goal is to predict the posted rate of a freight load from its lane, equipment, weight, date and market signals. The training file contains 48,000 loads from January to October 2025, covering 64 pickup and 64 delivery cities and three equipment types (Dry Van, Reefer, Flatbed). The validation set covers November–December 2025, so the task is a forecast into the future rather than a random sample of the same period. This shaped most of my decisions.

2 Data Cleaning

- **Weight:** 300 rows had missing weight and 292 had negative weight. Since weight cannot be missing or negative for a real load, I removed these rows.
- **Market index:** 374 rows had a missing `market_index`, which I also removed.
- **Load ID:** dropped, since it is just an identifier and carries no information about price.

After cleaning, 47,038 rows remained.

3 Train/Test Split

I used a **chronological split** instead of a random one: loads before 1 September 2025 are used for training (37,709 rows) and loads from September–October are held out for testing (9,329 rows). A random split would let the model see loads from the same days it is tested on, which leaks information about market conditions and gives an over-optimistic score. Since the real validation set lies entirely after the training data, testing on the last two months is the closest simulation of how the model will actually be used.

4 Features and Model

I used **CatBoost**, because the most important inputs (pickup, delivery, equipment) are categorical with many values, and CatBoost handles them natively without one-hot encoding. Distance, weight, coordinates, `market_index` and `quote_signal` are used as numeric features.

From the date I extracted `month` and `day_of_week` and dropped the raw date. I chose these after checking the data:

- **Day of week** shows a clear and consistent pattern: average rate per mile rises from \$2.17 on Monday to a peak of \$2.21 on Thursday and falls back to \$2.17 on Sunday.
- **Day of month** does not. I tested it, but its pattern does not repeat between months (average correlation of 0.07 between months), so it is noise. Including it caused an artificial \$30 drop in the middle of my December forecast, so I removed it.

Model settings: 250 iterations, learning rate 0.05, depth 8, RMSE loss.

5 Results

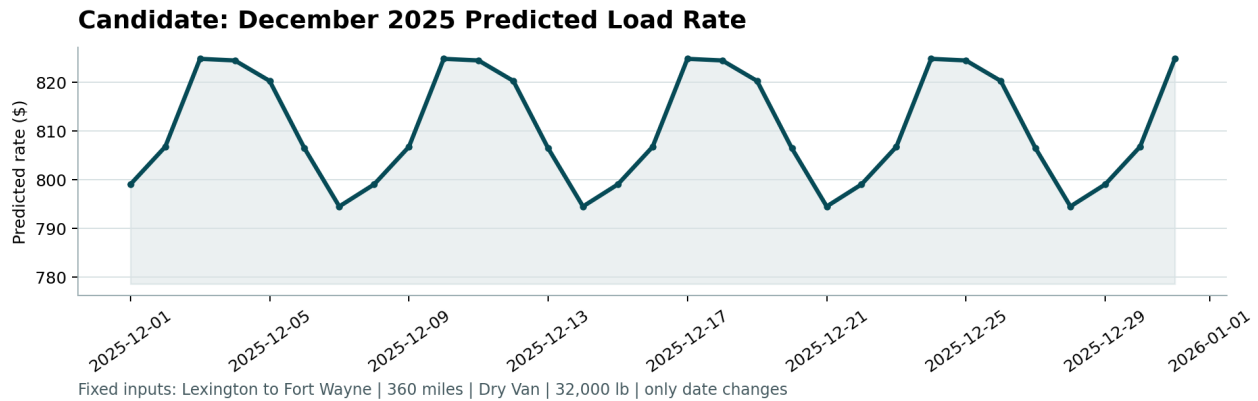
Test set, Sep–Oct 2025	Full model	December model
MAE	\$ 110.10	\$ 122.21
RMSE	\$ 638.37	\$ 639.07
MAPE	5.15 %	5.66 %

The full model (used for the validation predictions) uses all features. The December model (Section 6) only uses the inputs available in the December file. Removing `market_index` and `quote_signal` only increases MAE by about \$12, so most of the price is explained by the lane, distance, equipment and weight.

A typical prediction is off by about 5%. RMSE is much larger than MAE, which means most errors are small but a few are very large. These come from a small number of loads with implausible prices (see Limitations), which the model cannot predict and RMSE punishes heavily.

6 December Forecast

The December file only provides pickup, delivery, distance, equipment, weight and date — no `market_index` or `quote_signal`. I therefore trained a second CatBoost model using only these inputs plus `month` and `day_of_week`, so the model never depends on information that is not available at prediction time.



The forecast shows a repeating weekly cycle: rates peak midweek (around \$825) and are lowest on Sundays (around \$795). The weeks look the same because the only input that changes is the date, and the training data ends in October with a flat recent trend (monthly averages between \$2,340 and \$2,415 since July). The model has never seen December, so it does not invent holiday effects such as Christmas week that it has no data for. I think this is the honest forecast given the data.

7 Limitations and Next Steps

- **Cleaning vs. validation:** the validation set also contains missing and negative weights. Fixing them (e.g. taking the absolute value, since the negative values mirror the positive range) instead of dropping them would keep the training data closer to what the model sees at prediction time.
- **Outliers:** about 1.3% of training loads have implausible prices (below \$1 or above \$5 per mile). Removing them could make training more stable.

- **Tuning:** use early stopping and compare against a simple baseline (median rate per mile by equipment).
- **Holidays:** with more history including previous Decembers, holiday effects could be modelled.